HOOPS: Heuristic Optimization and Outcome Prediction System

Title: NBA Intelligent Performance & Decision Analytics Platform

Duration: 2 Months

Team Members

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Core Problem Statements

1. The "Siloed Data" Problem (Stats vs. Odds)

- The Problem: Currently, sports data is heavily fragmented into two separate silos. Team management and performance analysts look almost exclusively at standard game statistics (points, efficiency, rebounds), while the betting market focuses only on Vegas odds, implied probabilities, and point spreads.
- The Solution: The gap between these domains remains largely unbridged. The project builds a relational data pipeline to JOIN actual on-court performance metrics directly with financial betting markets. This allows for the mathematical evaluation of whether Vegas odds accurately reflect on-court reality, or if there is a systemic variance

that can be exploited using Supervised Machine Learning (specifically Logistic Regression and XGBoost classification models) to predict game outcomes.

2. The "Hardcoded Labels" Problem (Outdated Positions)

- The Problem: Utilizing traditional basketball positions (e.g., "Center", "Point Guard") as categorical features in a machine learning model is ineffective in the modern era. The game has evolved into a position-less format where a 7-foot-tall player might function exactly like a primary ball-handler. Using these hardcoded string labels introduces noise and causes predictive models to group players incorrectly.
- The Solution: Instead of relying on official roster titles, traditional positional labels are dropped entirely. The project utilizes Unsupervised Machine Learning (specifically K-Means Clustering) to dynamically group players into modern archetypes based on their actual statistical output and behavioral features on the court.

Defined ML Archetype Clusters (Target Output for Problem 2)

By processing behavioral features (detailed in the Schema section below), the K-Means clustering algorithm will classify all athletes into one of four modern, statistically verified archetypes:

1. Floor Generals: Primary playmakers and offensive engines characterized by high usage and high assist percentages.
2. Perimeter Snipers: Highly efficient outside scorers characterized by high true shooting percentages and 3-point attempt rates, but lower usage.
3. Paint Dominators: Interior forces characterized by high rebound percentages, free throw rates, and interior efficiency.

4. Defensive Enforcers: Low-usage role players characterized by high block and steal percentages, focusing primarily on defensive stops and hustle plays.

Project Objectives

1.Integrate multiple NBA Tables into a single analytics platform.
2.Perform Exploratory Data Analysis (EDA) to identify trends and patterns.
3.Analyze team and player performance using historical NBA data.
4.Build Machine Learning models to predict game outcomes and win probability.
5.Detect momentum shifts and identify key game-changing events.
6.Analyze betting odds, schedule difficulty, and team performance.
7.Create interactive Power BI dashboards for data visualization.
8.Generate actionable insights to improve coaching strategies and decision-making.

Dataset Description

The NBA Stats Dataset 1996 – Present (30 Years) consists of multiple interconnected datasets containing historical NBA game statistics, play-by-play events, player performance, team statistics, betting odds, and game schedules.

- Games Schedule:
  - Contains information about scheduled NBA games, including game ID, game date, season, home and away teams, final scores, winner, winning margin, and betting odds.
  - Useful for analyzing game outcomes, home-court advantage, and season performance.

- Games Index:
  - Stores historical NBA games from 1996 to the present
  - Includes season information, team details, scores, winners, game duration, and betting odds.
  - Used for long-term historical trend and team lifecycle analysis.
- Play-by-Play Data:
  - Contains detailed event-level information for every game, including scoring plays, turnovers, fouls, steals, rebounds, assists, substitutions, shot details, player actions, game clock, and periods.
  - Useful for momentum detection, win probability estimation, and event sequence analysis.
- Player Box Scores:
  - Stores detailed player statistics for every game, including minutes played, points, rebounds, assists, steals, blocks, shooting percentages, turnovers, plus/minus, and over 190 advanced performance metrics
  - Used for player performance analysis, impact evaluation, and machine learning.
- Team Box Scores:
  - Contains team-level statistics such as field goals, three-point shooting, rebounds, assists, turnovers, defensive performance, quarter-wise statistics, and more than 200 advanced team metrics.
  - Useful for team comparison, performance evaluation, and tactical analysis.
- Game Odds:
  - Stores historical betting odds including decimal odds, moneyline odds, odds update history, and betting source information.

• Useful for betting trend analysis, win probability estimation, and market expectation analysis.

Data Format

- The dataset is provided as SQLite database tables containing structured relational data with over 30 years of NBA history, millions of play-by-play events, and hundreds of statistical attributes.
- Suitable for Big Data Analytics, Exploratory Data Analysis (EDA), Machine Learning, Predictive Analytics, Time-Series Analysis, and Business Intelligence applications.

Core Engineering KPIs (Key Performance Indicators)

To quantify the success and analytical depth of the project, the following metrics are actively calculated and tracked:

1. Roster Style Balance: An optimization checker. It calculates whether the five active players on the court complement each other mathematically, preventing configurations that stack too many ball-dominant or redundant archetypes simultaneously.
2. Betting Line Variance: Calculates the absolute error of Vegas market predictions: abs(Actual Margin - Vegas Spread). High variance indicates that the market priced the game incorrectly, successfully identifying a quantifiable market inefficiency.
3. Archetype Dominance Tracker: Tracks how the ML-generated player clusters perform over time. It provides mathematical proof of how the statistical "meta" of the NBA has shifted across decades (e.g., isolating the exact year perimeter shooters began dominating scoring data).

4. Era-Adjusted Z-Score: Recognizes that a player's raw statistics from historical eras cannot be directly compared to modern metrics due to pace variations. This KPI applies a standard Z-score calculation (x - mean) / std_dev grouped by season_year, allowing for fair, standardized evaluation of athletes across different generations.
5. Quarterly Scorecard: Instead of evaluating only the final pts metric, quarter-by-quarter data (q1_pts to q4_pts) is parsed. This metric systematically flags franchises that consistently forfeit leads or execute significant comebacks during late-game pressure situations.
6. Dynamic Roster Synergy (Active Lineup Optimizer): Evaluates and optimizes the real-time mathematical balance of a team's current active 5-man lineup, explicitly preventing the stacking of redundant playstyles (e.g., fielding two high-usage "Floor Generals" simultaneously). The data pipeline dynamically filters the master statistical database to isolate only the current, active roster. This real-time data slice is passed into the trained K-Means algorithm's predict function. The system outputs an optimized rotation requiring a synergistic blend: 1 Floor General, 2 Perimeter Snipers, 1 Paint Dominator, and 1 Defensive Enforcer.

Required Data Schema & Feature Descriptions

Schema for Problem 1 (Market Variance & Team Stats)

To address the data fragmentation, the following features must be extracted and joined across the team_boxscores and game_odds tables:

- game_id *(Primary Key)*: The unique identifier required to execute the JOIN operation between the team performance table and the market odds table.
- team_abbreviation: The categorical identifier used to group, filter, and aggregate queries by specific franchises.

- pts & plus_minus: The actual points scored and the final point differential (margin of victory/defeat) of the game.
- spread: The Vegas-predicted point difference before the game started. Compared against plus_minus to calculate market variance.
- moneyline / home_odds / away_odds: The implied probability of a team winning outright. This serves as a baseline to evaluate against the internal Logistic Regression win probability outputs.
- over_under: The Vegas prediction for the total combined points in a matchup, used to track systemic over-scoring trends.
- off_rating & def_rating: Pace-adjusted metrics calculating points scored and allowed per 100 possessions. Provides the *true* mathematical efficiency of a team rather than raw point totals.
- wl *(Win/Loss)*: A clean binary target variable (W or L) utilized to train the classification models.

Schema for Problem 2 (Player Clustering & Archetypes)

To build an accurate K-Means clustering algorithm that identifies the modern archetypes, the following behavioral features are extracted from the player_boxscores table:

- player_name: The unique string identifier for the athlete.
- usg_pct *(Usage Percentage)*: A feature indicating the percentage of team plays an athlete is involved in while on the floor. Identifies ball-dominant assets versus role players.
- ts_pct *(True Shooting Percentage)*: A holistic metric measuring overall scoring efficiency by combining 2-pointers, 3-pointers, and free throws. Identifies highly efficient offensive players.

- ast_pct *(Assist Percentage)*: Used to identify the primary playmakers and offensive engines ("Floor Generals").
- reb_pct *(Rebound Percentage)*: Used to identify the primary interior forces and "Paint Dominators."
- fg3a *(3-Point Attempts)* or fg3_pct: The key separator for the "Perimeter Sniper" cluster, distinguishing outside shooters from interior scorers.
- blk_pct & stl_pct *(Block & Steal Percentage)*: Crucial for identifying the "Defensive Enforcer" archetype. Isolates athletes with low usage rates but high defensive output.
- tov_pct *(Turnover Percentage)* or ast_to: Refines the "Floor General" cluster by distinguishing primary ball handlers (who naturally incur turnovers) from catch-and-shoot players.
- season_year: Acts as the temporal filter. Querying the current year naturally drops all retired or historical players from the calculation.
- team_abbreviation: Acts as the categorical filter, allowing team management to isolate the prediction engine strictly to their own bench.
- fta_rate *(Free Throw Attempt Rate)*: Identifies slashers and physical interior players who frequently draw contact, separating them from perimeter-oriented athletes.

Tools & Technologies Used

- **Amazon S3 (Simple Storage Service):** Used as the central data lake to store raw, processed, and curated NBA datasets in a scalable and durable cloud environment.
- **AWS Lambda:** Used to automate data ingestion by triggering the transfer of NBA datasets from Kaggle into the Amazon S3 Raw bucket.

- **AWS Glue:** Used for serverless ETL (Extract, Transform, Load) to clean and transform raw NBA data. It also maintains the AWS Glue Data Catalog for metadata management.
- **Databricks Notebook:** Used as the development environment for writing, executing, and testing PySpark code. It enables interactive data exploration, feature engineering, machine learning model development, and distributed data processing.
- **PySpark:** Used for large-scale data processing, feature engineering, and building machine learning pipelines on NBA datasets.
- **Amazon Athena:** Used to execute serverless SQL queries directly on curated datasets stored in Amazon S3 using the AWS Glue Data Catalog.
- **Power BI:** Used to build interactive dashboards and visualize insights by connecting to Amazon Athena.

Architecture Diagram